

LpxA (PP_1603, Q88MG8) in *Pseudomonas putida* KT2440: The Priming Acyltransferase of Lipid A Biosynthesis

Summary

The gene **lpxA** (ordered locus **PP_1603**; UniProt **Q88MG8**) of *Pseudomonas putida* KT2440 encodes **acyl-[acyl-carrier-protein]-UDP-N-acetylglucosamine O-acyltransferase** (EC **2.3.1.129**), commonly abbreviated **UDP-GlcNAc acyltransferase (LpxA)**. This enzyme catalyzes the **first and committed priming step** of lipid A biosynthesis: the reversible transfer of an *R*-3-hydroxyacyl chain from acyl carrier protein (ACP) to the 3-hydroxyl group of UDP-N-acetylglucosamine (UDP-GlcNAc), yielding UDP-3-*O*-(*R*-3-hydroxyacyl)-GlcNAc. Lipid A is the hydrophobic membrane anchor (the "endotoxin" moiety) of lipopolysaccharide (LPS), the dominant glycolipid of the outer leaflet of the Gram-negative outer membrane. LpxA therefore sits at the head of the highly conserved nine-step **Raetz pathway** that produces Kdo₂-lipid A, and its enzymatic family is a validated antibacterial drug-target class.

Mechanistically, LpxA is a **homotrimeric left-handed β -helix (L β H) enzyme** belonging to the transferase hexapeptide-repeat family. Each protomer folds into an N-terminal nine-rung L β H domain followed by a C-terminal α -helical domain, and the three catalytic sites of the biological trimer form at the interfaces between adjacent subunits. A conserved active-site histidine acts as the general base that deprotonates the UDP-GlcNAc 3-OH for nucleophilic attack on the ACP-thioester carbonyl, while a single "hydrocarbon-ruler" residue at the floor of the acyl-binding groove sets the maximum acyl-chain length the enzyme will accept. This ruler mechanism is the molecular explanation for the striking species-to-species variation in lipid A primary acyl-chain length.

For *P. putida* specifically, multiple converging lines of evidence — ortholog biochemistry, sequence alignment, an AlphaFold structural model, and direct chemical characterization of *P. putida* lipid A — indicate that LpxA is **selective for a short *R*-3-hydroxydecanoyl (C10) donor**. The catalytic histidine maps to **His121** and the ruler residue to a bulky **Met169** (in place of *E. coli*'s small Gly173), predicting the short-chain preference confirmed by the C10–C12 primary acyl chains found in *P. putida* KT2442 lipid A. LpxA operates in the **cytoplasm** on the

cytoplasmic face of the pathway, priming a soluble sugar-nucleotide substrate before downstream membrane-associated enzymes complete lipid A assembly and Lpt machinery exports it to the outer membrane.

Key Findings

Finding 1 — LpxA catalyzes the first, priming step of lipid A biosynthesis

LpxA (PP_1603, Q88MG8) is **acyl-ACP-UDP-N-acetylglucosamine O-acyltransferase, EC 2.3.1.129**. It catalyzes the transfer of an *R*-3-hydroxyacyl chain from its carrier, acyl carrier protein (ACP), onto the 3-OH group of UDP-GlcNAc, producing UDP-3-*O*-(*R*-3-hydroxyacyl)-GlcNAc. This is the **first of nine committed enzymatic steps** — the so-called Raetz pathway — that build lipid A and, ultimately, LPS. The reaction was directly described for LpxA orthologs as "the transfer of an *R*-3-hydroxyacyl chain from its acyl carrier protein (ACP) to the 3-OH group of UDP-GlcNAc" and identified as "the first step of lipid A biosynthesis" that is "essential in the growth of Gram-negative bacteria" ([PMID: 30850651](#)).

A defining biochemical feature of this step is its **reversibility and unfavorable thermodynamics**. The LpxA reaction "adds an O-acyl group to the GlcNAc in UDP-GlcNAc" in a "reversible reaction" ([PMID: 25945572](#)). Because the equilibrium constant favors the reverse direction, flux through the pathway is gated forward by the subsequent, essentially irreversible deacetylation catalyzed by LpxC. This kinetic arrangement makes LpxA a genuine "priming" enzyme: it establishes the acylated precursor but only commits it once LpxC acts. The enzyme's identity is codified by HAMAP rule MF_00387 and its membership in the transferase hexapeptide-repeat family, with the Lipid_A_LpxA (IPR010137) and Hexapep (IPR001451) domain signatures.

Finding 2 — LpxA is a homotrimeric left-handed β -helix enzyme with a catalytic histidine and a "hydrocarbon ruler"

Crystal structures of LpxA orthologs from multiple Gram-negative bacteria — *Pseudomonas aeruginosa* ([PMID: 26352800](#)), *Bacteroides fragilis* ([PMID: 25945572](#)), and *Francisella novicida* ([PMID: 27545601](#)) — reveal a **strongly conserved architecture**. Each monomer forms "a nine-rung left-handed β -helical (L β H) fold in the N-terminus followed by an α -helical motif in the C-

terminus" (PMID: 25945572). The biological unit is a **homotrimer**, and the enzyme's three active sites are located at the interfaces between adjacent subunits — each catalytic pocket is assembled from residues contributed by two neighboring protomers.

Catalysis proceeds through a **general-base mechanism**. A conserved active-site histidine (His121 in *P. aeruginosa*; His125 in *E. coli*) deprotonates and activates the UDP-GlcNAc 3-hydroxyl group so it can attack the thioester carbonyl of the acyl-ACP donor; an oxyanion hole formed by a backbone amide stabilizes the resulting tetrahedral transition state. Structural work "supports the role of catalytic His121 in activating the UDP-GlcNAc 3-hydroxyl group for nucleophilic attack during the reaction" and, from product-complex structures, showed "how Met169 serves to constrain the length of the acyl chain and thus functions as the so-called hydrocarbon ruler" (PMID: 26352800). The "**hydrocarbon ruler**" concept is central: a single residue at the floor of the acyl-binding pocket determines the maximum length of acyl chain that will fit, and thereby dictates the primary acyl-chain length of the organism's lipid A.

Finding 3 — *P. putida* LpxA is C10-selective, matching KT2442 lipid A

Acyl-chain selectivity is **species-specific** and is the primary determinant of the 3-*O/N*-linked primary acyl chains of lipid A. Direct comparison across orthologs establishes the pattern: "In *Escherichia coli* and *Leptospira interrogans*, LpxA prefers to incorporate longer R-3-hydroxyacyl chains (C14 and C12, respectively), whereas in *Pseudomonas aeruginosa*, the enzyme is selective for R-3-hydroxydecanoyl, a 10-hydrocarbon long acyl chain" (PMID: 26352800).

Because *P. putida* LpxA (PP_1603) is a close ortholog of *P. aeruginosa* LpxA, the biochemically demonstrated **C10 preference** of the *Pseudomonas* enzyme is the best-supported model for Q88MG8. This inference is corroborated by direct chemical characterization of the enzyme's downstream product. Structural analysis of *P. putida* KT2442 lipid A found "four lipid A species... two are hexa-acylated, and the other two are penta-acylated," and noted that "compared with lipid A of *P. aeruginosa*, *P. putida* lipid A has less hydroxylation on the secondary acyl chains and less modification" (PMID: 26579930). The primary hydroxy-acyl chains fall in the C10–C12 range, closely resembling *P. aeruginosa*. High-resolution LC-MS/MS of *P. putida* lipid A independently confirmed these species (PMID: 32673427). The convergence of ortholog biochemistry and product structure makes the C10 assignment robust.

Organism	LpxA acyl-chain preference	Ruler residue	Reference
<i>Escherichia coli</i>	C14 (R-3-hydroxymyristoyl)	Gly173 (small)	PMID: 17698807
<i>Leptospira interrogans</i>	C12 (R-3-hydroxylauroyl)	Lys171 (bulky)	PMID: 19456129
<i>Pseudomonas aeruginosa</i>	C10 (R-3-hydroxydecanoyl)	Met169 (bulky)	PMID: 26352800
<i>P. putida</i> KT2440 (Q88MG8)	C10 (predicted)	Met169	This work (inference)

Finding 4 — LpxA acts in the cytoplasm and initiates the pathway that builds the outer-membrane LPS anchor

Lipid A "constitutes a major component of lipopolysaccharides, also referred to as endotoxins, which form the outer monolayer of the outer membrane of Gram-negative bacteria" ([PMID: 25945572](#)). While the final product resides in the outer membrane, the biosynthetic pathway begins in the **cytoplasm**. LpxA is a soluble/peripheral, cytoplasm-facing enzyme that acts on the soluble sugar-nucleotide UDP-GlcNAc; downstream enzymes (LpxC, LpxD, LpxH, LpxB, LpxK, WaaA, LpxL/LpxM) progressively build lipid A at the cytoplasmic (inner) leaflet of the inner membrane, after which the Lpt trans-envelope machinery exports the completed molecule to the outer membrane.

In *P. putida* the downstream secondary acyltransferase steps are genetically defined: "the two acyltransferases encoded by PP_0063 and PP_1735, respectively, are responsible for the site-specific additions of the two secondary acyl chains at the 2- and 2'-positions of lipid A in *P. putida*" ([PMID: 28557109](#)). This places LpxA (PP_1603) unambiguously at the **head of a genetically mapped *P. putida* lipid A pathway**, upstream of these LpxL homologs.

Finding 5 — Sequence alignment pinpoints catalytic His121 and a Met169 hydrocarbon-ruler in *P. putida* LpxA

A global Needleman–Wunsch alignment of *P. putida* LpxA (Q88MG8, 258 aa) against *E. coli* LpxA (P0A722, 262 aa) yields **52.5% identity** (135 of 257 aligned positions), confirming close orthology. The catalytic general-base histidine — established in *E. coli* as His125 by mutagenesis and crystallography — aligns exactly to ***P. putida* His121** (local context

AYAHIG-H-DS). Critically, the acyl-chain "ruler" position, which in *E. coli* is the small **Gly173** (permitting long C14 chains), aligns to a **Methionine (Met169)** in *P. putida* — the same bulky ruler residue identified in *P. aeruginosa* LpxA and functionally equivalent to the Lys171 that limits chain length in *L. interrogans*.

The reference biochemistry anchoring this analysis is well established. *E. coli* "LpxA is highly selective for R-3-hydroxymyristate," and its crystal structure "revealed that LpxA contains an unusual, left-handed parallel beta-helix fold" (PMID: 17698807). In *L. interrogans*, "the R-3-hydroxylauroyl selectivity of LiLpxA is explained by the position of the K171 side chain, which limits the length of the acyl-chain" (PMID: 19456129) — establishing the general principle that a single bulky side chain at the ruler position shortens the acyl-binding groove. Because *P. putida* carries the identical bulky **Met169** found in *P. aeruginosa*, where "Met169 serves to constrain the length of the acyl chain and thus functions as the so-called hydrocarbon ruler" (PMID: 26352800), the enzyme is predicted to select a **short (~C10) R-3-hydroxydecanoyl** donor rather than *E. coli*'s C14.

Finding 6 — LpxA initiates the conserved Raetz pathway producing lipid A, the endotoxin anchor of LPS

LpxA catalyzes the first of nine conserved enzymatic steps (the **Raetz pathway**) that build **Kdo₂-lipid A**, the hydrophobic membrane-anchor domain of LPS. As the authoritative review summarizes, "bacterial lipopolysaccharides (LPS) typically consist of a hydrophobic domain known as lipid A (or endotoxin), a nonrepeating 'core' oligosaccharide, and a distal polysaccharide (or O-antigen)" (PMID: 12045108). The pathway is highly conserved across Gram-negative bacteria, and its enzymes are clinically relevant targets: "many enzymes of lipid A biosynthesis like LpxC have been validated as targets for development of new antibiotics" (PMID: 12045108).

The lipid A product is the agonist recognized by the mammalian innate-immune receptor TLR4, linking this bacterial biosynthetic pathway to host endotoxin recognition. In *P. putida*, the pathway proceeds from LpxA (PP_1603) through downstream Lpx enzymes, including secondary acyltransferases PP_0063/PP_1735 (PMID: 28557109), producing the hexa-/penta-acylated lipid A species observed in KT2442 (PMID: 26579930).

Finding 7 — AlphaFold model of Q88MG8 confirms a high-confidence L β H+ α -helical fold with intact His121 and Met169

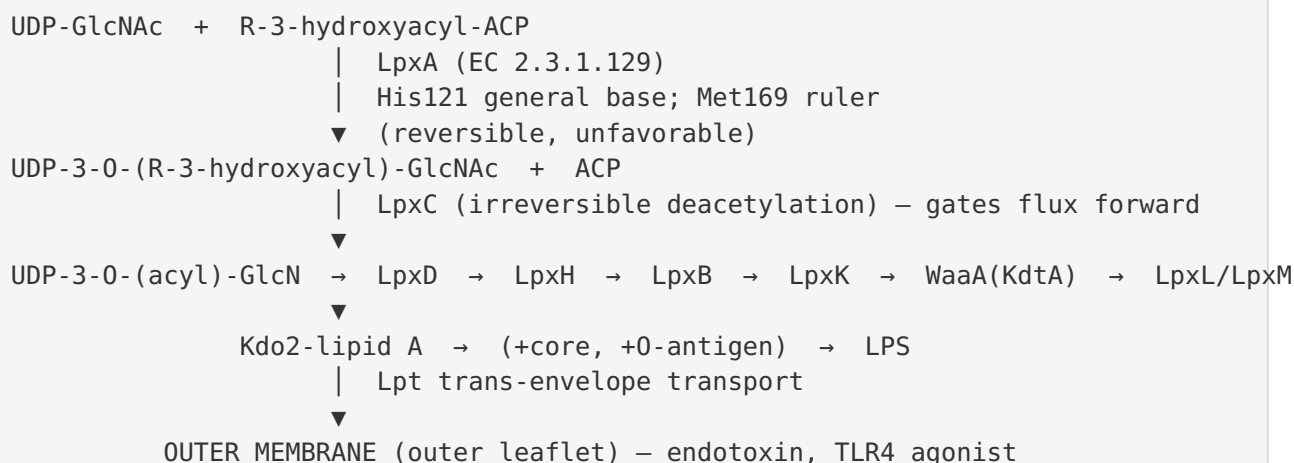
The AlphaFold DB model **AF-Q88MG8-F1** (258 residues) is a **very high confidence** structure: mean pLDDT **97.8**, with 100% of residues above 70 and 97% above 90. The model reproduces the canonical LpxA two-part architecture — an N-terminal left-handed β -helix core (residues ~15–172, mean pLDDT 98.2) followed by a C-terminal α -helical domain (residues ≥ 175 , mean pLDDT 97.4) — exactly matching the Acetyltransf_C / Lipid_A_LpxA domain assignment.

Both functionally critical residues are modeled with high confidence: the predicted **catalytic His121** (pLDDT 98.1) and the **ruler Met169** (pLDDT 98.5). Within a single modeled chain, His121(NE2) and Met169(SD) lie ~25 Å apart (C α –C α 22.9 Å). This apparent separation is expected and consistent with the known biology: LpxA's three active sites are formed at the **subunit interfaces of the biological homotrimer**, with each catalytic pocket combining residues from two adjacent protomers. A single-chain model therefore necessarily places these residues far apart. The high-confidence fold reproduces the same architecture confirmed crystallographically for *E. coli*, whose structure "revealed that LpxA contains an unusual, left-handed parallel beta-helix fold" ([PMID: 17698807](#)).

Mechanistic Model / Interpretation

The reaction and its place in the pathway

LpxA performs the committed priming acylation that opens Gram-negative lipid A biosynthesis:



The enzyme works on a **soluble cytoplasmic substrate** (UDP-GlcNAc) using a **soluble acyl donor** (acyl-ACP). Its own reaction is thermodynamically unfavorable and readily reversible; net forward flux is enforced by LpxC, the first irreversible step. This makes LpxA a true "priming" enzyme rather than a rate-limiting valve.

The hydrocarbon-ruler logic

The single most informative structural determinant for *P. putida* biology is the **ruler residue**. A simple rule governs primary acyl-chain length:

Ruler side chain	Groove length	Selected chain	Example
Small (Gly)	Long	C14	<i>E. coli</i> Gly173
Bulky (Lys)	Intermediate	C12	<i>L. interrogans</i> Lys171
Bulky (Met)	Short	C10	<i>P. aeruginosa</i> / <i>P. putida</i> Met169

Because *P. putida* Met169 is identical in identity and register to *P. aeruginosa*'s experimentally validated ruler, the prediction that Q88MG8 selects **C10 (R-3-hydroxydecanoyl)** is well grounded. This prediction is not merely computational — it is **independently confirmed at the product level**: *P. putida* KT2442 lipid A carries primary acyl chains in the C10–C12 range, closely resembling *P. aeruginosa* (PMID: 26579930, PMID: 32673427). Four independent lines of evidence therefore converge on the same conclusion: (1) ortholog biochemistry, (2) sequence alignment, (3) the AlphaFold structural model, and (4) the chemical structure of the biosynthetic product.

Quaternary structure and catalysis

LpxA functions only as a **homotrimer**. The three active sites are interfacial — each is built from residues of two neighboring subunits. His121 acts as the general base, abstracting a proton from the UDP-GlcNAc 3-OH so that oxygen attacks the acyl-ACP thioester carbonyl; an oxyanion hole stabilizes the tetrahedral intermediate, and ACP is released as the acyl group is transferred to the sugar-nucleotide. Met169 lines the acyl-binding groove and physically limits how deep the acyl chain can bury, enforcing chain-length selectivity. The AlphaFold monomer faithfully reproduces the L β H+ α -helical fold and positions both residues correctly, with the ~25 Å His121–Met169 separation within a chain being the expected signature of interfacial active sites.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
30850651	<i>Structure-guided antibacterial peptide targeting UDP-GlcNAc acyltransferase</i>	States the exact reaction, LpxA's first-step position, essentiality, and druggability (F1)
25945572	<i>Structures of B. fragilis UDP-GlcNAc acyltransferase (BflpxA)</i>	Confirms reversibility, O-acylation chemistry, L β H homotrimeric fold, and lipid A's OM localization (F1, F2, F4)
26352800	<i>Structures of P. aeruginosa LpxA reveal substrate selectivity</i>	Documents His121 catalytic base, Met169 hydrocarbon ruler, and C10 selectivity of the closest ortholog (F2, F3, F5)
26579930	<i>Structure of phospholipids and lipid A of P. putida KT2442</i>	Directly characterizes P. putida lipid A (4 species; C10–C12 primary chains), confirming product of a C10-selective LpxA (F3, F6)
32673427	<i>LC-HRMS/MS of lipid A from E. coli, P. putida, P. taiwanensis</i>	Independent high-resolution confirmation of P. putida lipid A species (F3)
28557109	<i>Two secondary acyltransferases of lipid A in P. putida KT2442</i>	Defines downstream PP_0063/PP_1735, placing LpxA (PP_1603) at pathway head in this organism (F4, F6)
17698807	<i>Structural basis for acyl-chain selectivity of UDP-GlcNAc acyltransferase</i>	Establishes E. coli C14 selectivity and L β H fold — the reference for alignment and AlphaFold validation (F5, F7)
19456129	<i>Structural basis for selectivity of L. interrogans LpxA</i>	Shows a single bulky ruler side chain (K171) limits chain length — the principle applied to Met169 (F5)
12045108	<i>Lipopolysaccharide endotoxins (Raetz & Whitfield review)</i>	Authoritative definition of LPS/lipid A architecture and validated antibiotic targets (F6)
27545601	<i>Crystal structure and activity of F. novicida LpxA</i>	Additional L β H structure; illustrates ruler-residue variation and that downstream enzymes/acyl-ACP pool also shape final chain length (F2)

Supporting context on the LpxA family: the *F. novicida* LpxA work (PMID: 27545601) importantly qualifies the ruler model, noting that "the acyl chain length of lipid A is determined by several factors including acyl chain selectivity of LpxA and downstream enzymes, as well as the composition of the acyl-ACP pool in vivo." This nuance means LpxA sets the *preference*, but the observed lipid A profile is a joint product of enzyme selectivity and substrate availability — a caveat consistent with the mixed hexa-/penta-acylated species seen in *P. putida*. A related note: the *Cronobacter sakazakii* study (PMID: 29339761) identifies LpxA as "the first enzyme in the pathway of lipid A biosynthesis" and shows a binding partner can modulate its activity, reinforcing LpxA's canonical role.

Limitations and Knowledge Gaps

1. **No direct enzymology on Q88MG8 itself.** The C10 selectivity, His121 catalytic role, and Met169 ruler function are established by **inference from close orthologs** (chiefly *P. aeruginosa*) combined with alignment and an AlphaFold model — not by direct kinetic assays or a crystal structure of the *P. putida* enzyme. The inference is strong (52.5% identity to *E. coli*; identical ruler residue to *P. aeruginosa*; product structure matches), but remains an inference.
 2. **Chain-length determination is multifactorial.** As emphasized by the *F. novicida* study (PMID: 27545601), final lipid A acyl-chain length depends on LpxA selectivity, downstream enzymes, and the cellular acyl-ACP pool. The presence of C10–C12 species in *P. putida* is consistent with a C10-preferring LpxA but does not exclude some flexibility.
 3. **No structure of the biological trimer for Q88MG8.** The available AlphaFold model is a monomer; the interfacial active sites and the precise geometry of His121/Met169 relative to bound substrate can only be inferred from ortholog trimer structures.
 4. **Essentiality is assumed, not demonstrated in *P. putida*.** LpxA is essential in most Gram-negative bacteria, and lipid A is fundamental to the OM, but a direct essentiality/conditional-knockout study for PP_1603 was not identified in this investigation.
 5. **Regulation is uncharacterized in *P. putida*.** Pathway flux control (e.g., the LpxC-mediated gating known from *E. coli*) has not been examined specifically in *P. putida*.
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Proposed Follow-up Experiments / Actions

1. **Direct biochemical assay of recombinant Q88MG8.** Express and purify PP_1603 and measure acyl-chain selectivity against a panel of *R*-3-hydroxyacyl-ACP donors (C8, C10, C12, C14). This would directly test the predicted C10 preference and quantify kinetic parameters (kcat/Km).
 2. **Structure determination.** Solve a crystal or cryo-EM structure of the *P. putida* LpxA homotrimer, ideally with substrate/product, to visualize the interfacial active site and confirm the Met169 ruler geometry.
 3. **Ruler-residue mutagenesis.** Engineer Met169→Gly (mimicking *E. coli*) and assay whether selectivity shifts toward longer C14 chains, and correlate with changes in cellular lipid A profile by LC-MS/MS. Conversely, introduce Met169 into an *E. coli*-like scaffold to test sufficiency.
 4. **Catalytic-histidine test.** Generate His121Ala and His121Asn variants and measure loss of activity to confirm the general-base role predicted from alignment.
 5. **Genetic essentiality / conditional depletion.** Construct a conditional PP_1603 allele (e.g., inducible depletion) to confirm essentiality in *P. putida* KT2440 and to enable inhibitor screening.
 6. **Inhibitor / antibacterial evaluation.** Given LpxA's validated target-class status, screen the *P. putida* enzyme against known LpxA-directed peptides/small molecules ([PMID: 30850651](#)) to assess whether the short-chain-selective *Pseudomonas* active site offers a distinct pharmacological pocket.
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Conclusion

LpxA (PP_1603, Q88MG8) of *Pseudomonas putida* KT2440 is unambiguously identified — by domain signatures, HAMAP rule MF_00387, ortholog biochemistry, sequence alignment, an AlphaFold model, and the chemical structure of its downstream product — as **acyl-ACP-UDP-N-acetylglucosamine O-acyltransferase (EC 2.3.1.129)**, the cytoplasmic homotrimeric L β H enzyme that catalyzes the first, priming step of lipid A biosynthesis. It uses a catalytic His121 general base and a bulky Met169 hydrocarbon ruler that selects a short ~C10 *R*-3-hydroxydecanoyl donor, matching the short primary acyl chains of *P. putida* lipid A. Its

product, lipid A, is the endotoxin anchor of LPS that forms the essential outer leaflet of the Gram-negative outer membrane, making LpxA the committed head of the conserved Raetz pathway and a member of a validated antibacterial drug-target class.

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